Feature Report: Behavioral Features

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Basic Information

- **Total Customers**: 61,410
- **Total Features**: 27
- **Feature Columns**: total_volume, avg_transaction_amount, max_transaction_amount, transaction_amount_std, amount_to_income_ratio, amount_to_sales_ratio, amount_to_revenue_ratio, lifestyle_mismatch, severe_lifestyle_mismatch, has_missing_income, has_missing_sales, amount_cv, high_amount_variability, debit_volume, credit_volume, net_flow, debit_credit_ratio, high_debit_ratio, max_txn_to_income_ratio, max_txn_to_sales_ratio, max_txn_to_revenue_ratio, single_txn_exceeds_revenue, txn_count_total, transaction_consistency, consistent_pattern_flag, behavioral_risk_score, high_behavioral_risk

Data Quality Checks

☑ No Null Values

All features have complete data.

☑ No Duplicate Customer IDs

Feature Statistics

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero %
total_volume	Float	0.00	2677539042.00	6120900.64	2311184.50	28812801.85	6	0.01%
avg_transaction_amount	Float	0.00	224838493.00	98145.85	35126.97	1135873.73	6	0.01%
max_transaction_amount	Float	0.00	2051810761.00	2248623.79	350000.00	15858254.32	6	0.01%
transaction_amount_std	Float	0.00	483604743.40	301463.30	66953.72	2721253.14	702	1.14%
amount_to_income_ratio	Float	0.00	641237.23	266.81	0.22	3365.03	22,992	37.44%
amount_to_sales_ratio	Float	0.00	453407871.00	58007.52	0.00	3101551.92	57,379	93.44%
amount_to_revenue_ratio	Float	0.00	453407871.00	58274.34	0.35	3101548.75	18,961	30.88%
lifestyle_mismatch	Integer	0	1	0.22	0	0.41	48,063	78.27%
severe_lifestyle_mismatch	Integer	0	1	0.17	0	0.37	51,072	83.17%
has_missing_income	Integer	0	1	0.24	0	0.43	46,732	76.10%
has_missing_sales	Integer	0	1	0.07	0	0.25	57,132	93.03%
amount_cv	Float	0.00	10.00	2.45	2.09	1.54	702	1.14%
high_amount_variability	Integer	0	1	0.54	1	0.50	28,297	46.08%
debit_volume	Float	0.00	1437637276.00	2861381.83	886504.50	14983096.02	550	0.90%
credit_volume	Float	0.00	2052481891.00	3259518.81	1213366.50	18248172.39	2,080	3.39%
net_flow	Float	-628883171.00	2052309612.00	398136.98	191413.50	16876029.55	6	0.01%
debit_credit_ratio	Float	0.00	10.00	1.54	0.70	2.29	2,624	4.27%
high_debit_ratio	Integer	0	1	0.19	0	0.39	49,627	80.81%
max_txn_to_income_ratio	Float	0.00	630987.45	98.18	0.04	3157.55	22,992	37.44%
max_txn_to_sales_ratio	Float	0.00	299434000.00	23841.46	0.00	1512651.10	57,379	93.44%
max_txn_to_revenue_ratio	Float	0.00	299434000.00	23939.64	0.06	1512652.85	18,961	30.88%
single_txn_exceeds_revenue	Integer	0	1	0.19	0	0.39	49,939	81.32%
txn_count_total	Integer	1	4,991	96.13	63	102.92	0	0.00%

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero %
transaction_consistency	Float	0.00	1.00	0.33	0.32	0.12	6	0.01%
consistent_pattern_flag	Integer	0	0	0.00	0	0.00	61,410	100.00%
behavioral_risk_score	Integer	0	9	2.35	1	2.66	12,844	20.92%
high_behavioral_risk	Integer	0	1	0.25	0	0.44	45,809	74.60%

Validation Checks

☑ No Unexpected Negative Values

☑ No Infinite Values

Outlier Detection (Values > 3 Standard Deviations)

- total_volume: 510 outliers (0.83%)
- avg_transaction_amount: 99 outliers (0.16%)
- max_transaction_amount: 361 outliers (0.59%)
- transaction_amount_std: 225 outliers (0.37%)
- amount_to_income_ratio: 141 outliers (0.23%)
- amount_to_sales_ratio: 66 outliers (0.11%)
- amount_to_revenue_ratio: 66 outliers (0.11%)
- has_missing_sales: 4,278 outliers (6.97%)
- amount_cv: 1,721 outliers (2.80%)
- debit_volume: 544 outliers (0.89%)

... and 8 more features with outliers

Flag Feature Distributions

Feature	True Count	True Percentage	False Count	False Percentage
consistent_pattern_flag	0	0.00%	61,410	100.00%

Summary

- ☑ **Total Features**: 27
- ☑ **Data Completeness**: 3.70%
- ☑ **Customers Covered**: 61,410
- ☑ **Flag Features**: 1

Report generated automatically by feature engineering pipeline